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HW02 – Domain Testing on EShop

Information

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Course	CS423 / CSC13003 – Software Testing

Assignment	HW02
Submission Date	29/06/2026
Submission Deadline	29/06/2026
AI Tools Used	GitHub Copilot, Antigravity
Pandoc Version	3.9.0.2

FR-06

Domain Testing

Domain Testing Process:

- 1. Identify input variables:** For the product detail feature, two key input variables affecting business logic are:
 - `product_id` (parameter passed in the URL).
 - `quantity` (input field for product quantity to add to cart).
- 2. Determine domain rules and Equivalence Partitioning:**
 - For `product_id`:
 - Valid partition: Product ID is a positive integer and exists in the database.
 - Invalid partition: ID is a positive integer but does not exist in the DB, or ID is not a number.
 - For `quantity`:
 - Valid partition: Positive integer greater than or equal to 1.
 - Invalid partition: Equal to 0, negative number, decimal number, left blank, or containing non-numeric characters.
- 3. Select representative values:** Select specific values within each partition to design test cases.
- 4. Establish test cases:** Use the Single Fault Assumption to combine input values. A total of 9 test cases from TC-PRODUCT-DETAIL-001 to TC-PRODUCT-DETAIL-009 were designed.

Input Domain Analysis:

Variable 1: product_id (URL Parameter)

Partition Code	Partition Type	Partition Description	Representative Value
EP-ID-1	Valid	Product ID exists in the system	1
EP-ID-2	Invalid	ID is a number but does not exist in DB	9999
EP-ID-3	Invalid	ID is not an integer (contains letters)	" abc "

Variable 2: quantity (Add to cart quantity field)

Partition Code	Partition Type	Partition Description	Representative Value
EP-Q-1	Valid	Positive integer >= 1	2
EP-Q-2	Invalid	Value is 0	0
EP-Q-3	Invalid	Negative integer	- 1
EP-Q-4	Invalid	Decimal number	1 . 5
EP-Q-5	Invalid	Empty data field	" "
EP-Q-6	Invalid	String of letters or special characters	" abc "

List of Test Cases Designed by Domain Testing:

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
TC-PRODUCT-DETAIL-001	Access product details with a valid ID	EP-ID-1, EP-Q-1 (default)	Display full product details (image, name, price, category, description)
TC-PRODUCT-DETAIL-002	Access product details with a non-existent ID (9999)	EP-ID-2	Display 404 error page or appropriate error message
		EP-ID-3	Return an error page or redirect to homepage

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
TC-PRODUCT-DETAIL-003	Access product details with a non-numeric ID ("abc")		
TC-PRODUCT-DETAIL-004	Add product to cart with valid quantity (2)	EP-Q-1, EP-ID-1	Product added to cart; show success notification and update cart badge
TC-PRODUCT-DETAIL-005	Add product to cart with quantity = 0	EP-Q-2, EP-ID-1	System rejects, errors with invalid quantity
TC-PRODUCT-DETAIL-006	Add product to cart with negative quantity (-1)	EP-Q-3, EP-ID-1	System rejects, displays error message
TC-PRODUCT-DETAIL-007	Add product to cart with decimal quantity (1.5)	EP-Q-4, EP-ID-1	System rejects, only accepts positive integers
TC-PRODUCT-DETAIL-008	Add product to cart with empty quantity	EP-Q-5, EP-ID-1	System displays error and prompts for valid quantity
TC-PRODUCT-DETAIL-009	Add product to cart with non-numeric quantity ("abc")	EP-Q-6, EP-ID-1	System rejects and displays input data error

Boundary Value Analysis (BVA)

Boundary Value Analysis (BVA) Process:

- 1. Identify boundary variables:** The `quantity` variable has a clear lower bound according to the requirements specification (must be ≥ 1). The `product_id` variable is an identifier in the database, so BVA does not apply.
- 2. Identify boundary test points (3-value BVA):**
 - With a minimum boundary $min = 1$ and step size of 1 (integer):
 - **OFF** ($min - 1$) = 0 (Invalid value located just outside the boundary).
 - **ON** (min) = 1 (Valid value located exactly on the boundary).
 - **IN** ($min + 1$) = 2 (Valid value located just inside the boundary).
- 3. Map and supplement test cases:**
 - The OFF point (0) has been tested in TC-PRODUCT-DETAIL-005.
 - Add TC-PRODUCT-DETAIL-010 for the ON point (1).

- Add TC - PRODUCT - DETAIL - 011 for the IN point (2).

Boundary Value Analysis Table:

- **Boundary B1:** Minimum limit of the quantity field (≥ 1).
- **Boundary Type:** Closed minimum boundary.
- **Boundary Value:** 1.
- **Step Size:** 1.

Boundary Test Point	Test Value	Expected Result	Linked Test Case
OFF (min - 1)	0	Rejected and displays error	TC-PRODUCT-DETAIL-005
ON (min)	1	Accepted and successfully added	TC-PRODUCT-DETAIL-010
IN (min + 1)	2	Accepted and successfully added	TC-PRODUCT-DETAIL-011

List of Test Cases Designed by BVA:

Test Case ID	Test Case Description	Boundary Point	Value	Expected Result
TC-PRODUCT-DETAIL-005	Add product to cart with quantity = 0	OFF	0	Rejected (cannot add to cart)
TC-PRODUCT-DETAIL-010	Add product to cart with quantity = 1	ON	1	Accepted and successfully added to cart
TC-PRODUCT-DETAIL-011	Add product to cart with quantity = 2	IN	2	Accepted and successfully added to cart

AI Gap Analysis

During the process of using AI tools (GitHub Copilot and Antigravity) to design test cases for the Product Detail feature (FR-06), several shortcomings and logic errors on the AI’s part were identified:

- **Equivalence Partitioning logic error:** The AI initially defined the valid partition EP-Q-1 for the quantity to add to the cart as > 1. However, according to the requirements specification, the minimum quantity allowed to add is 1 (i.e., >= 1). Defining it as > 1 missed the crucial boundary ON point of quant i t y = 1.
- **Missing test case implementation:** The AI declared the invalid partition EP-Q-6 (quantity containing alphabetic characters " abc ") in the analysis table, but completely missed designing any test case to execute this partition in the initial test case list.
- **Vague expected results:** For the decimal quantity test case (quant i t y = 1 . 5), the AI provided a vague expected result: “rejected OR rounded”. In contrast, the system specification only accepts positive integers, so the correct expected result must be complete rejection.
- **Duplicate test cases between DT and BVA:** The boundary OFF test case (quant i t y = 0) was generated by the AI as a new BVA test case, which completely duplicated the domain testing case TC-PRODUCT-DETAIL-005.
- **Omission of integration and security requirements:** The AI missed navigation test scenarios such as Breadcrumb (FR-23), SEO/UI attribute tests like Image alt text (FR-24), scenarios for adding to cart when not logged in, and security testing against XSS attacks (FR-21/SEC-04) when displaying a product name containing malicious HTML tags.

Root cause: The AI designed based on heuristics and common templates of standard product detail pages rather than actually reading, understanding, and strictly adhering to the specific project specifications. The AI also had difficulty maintaining consistency between the variable declaration tables and the actual test case lists.

Bug Report

B u g I D	Title	Se ver ity	Detecting Test Case	GitHub Link
B U G -0 6-	[BUG][Product Detail] - Missing Category Name	Ma jor / P1	TC-PRODUCT-DETAIL-001	Issue #62

B u g I D	Title	Se ver ity	Detecting Test Case	GitHub Link
01				
B U G -06-02	[BUG][Product Detail] - Must double-click to 'add to cart'; No badge/toast indicates the 'add to cart' action	Ma jor / P1	TC-PRODUCT-DETAIL-004	Issue #63
B U G -06-03	[BUG][Product Detail] - Product with invalid quantity still be added to cart	Ma jor / P1	TC-PRODUCT-DETAIL-005 / TC-PRODUCT-DETAIL-006 / TC-PRODUCT-DETAIL-007 / TC-PRODUCT-DETAIL-008 / TC-PRODUCT-DETAIL-009	Issue #64
B U G -06-04	[BUG][Product Detail] - Un-auth user can add to cart	Ma jor / P1	TC-PRODUCT-DETAIL-012	Issue #65
B U G -0	[BUG][Product Detail] - Missing breadcrumb	Mi nor / P2	TC-PRODUCT-DETAIL-013	Issue #66

B u g I D	Title	Se ver ity	Detecting Test Case	GitHub Link
6- 0 5				

BUG-06-01: [BUG][Product Detail] - Missing Category Name

- **Issue ID:** #62
- **Severity / Priority:** Major / P1
- **Test Case:** TC-PRODUCT-DETAIL-001
- **Environment:** Chrome, macOS, http://localhost:5173/product/1, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 - 1. Open the Product Detail page with a valid product ID.
 - 2. Observe the category name.
- **Expected Result:** Display the category name fully.
- **Actual Result:** Missing product category name.

yuran1811 / hcmus-sw-testing--eshop-sut

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📊 Insights

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[BUG][Product Detail] - Missing Category Name #62

New issue

🔍 Open

yuran1811

opened 6 hours ago

Last edited by yuran1811

Owner

Found by Test Case

TC-PRODUCT-DETAIL-001

Requirement liên quan

FR-06

Severity / Priority

Major / P1

Environment

Chrome, macOS, http://localhost:5173/product/1_85af3ba

Steps to reproduce

1. Mở trang Product Detail với product ID hợp lệ.

2. Quan sát tên danh mục.

Expected result

Hiển thị đầy đủ tên danh mục.

Actual result

Thiếu tên danh mục của sản phẩm.

Evidence

EShop

Giỏ hàngChân, Test UserThoát

iPhone 15

iPhone 15 Pro Max

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Điện thoại cao cấp của Apple

Số lượng: 1

Thêm vào giỏ hàng

Assignees

No one - [Assign yourself](#)

Labels

module:product

priority:P1

severity:major

status:new

type:bug

Projects

eshop-sut

Status

To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Evidence:

BUG-06-02: [BUG][Product Detail] - Must double-click to 'add to cart'; No badge/toast indicates the 'add to cart' action

- **Issue ID:** #63
- **Severity / Priority:** Major / P1
- **Test Case:** TC-PRODUCT-DETAIL-004
- **Environment:** Chrome, macOS, <http://localhost:5173/product/1>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Open the Product Detail page.
 2. Enter a valid quantity.
 3. Click "Add to Cart" once.
- **Expected Result:** Product added immediately and navbar has a quantity badge.
- **Actual Result:** Must double-click to add, and navbar does not display the quantity badge.

BUG-06-03: [BUG][Product Detail] - Product with invalid quantity still be added to cart

- **Issue ID:** #64
- **Severity / Priority:** Major / P1
- **Test Case:** TC-PRODUCT-DETAIL-005 / TC-PRODUCT-DETAIL-006 / TC-PRODUCT-DETAIL-007 / TC-PRODUCT-DETAIL-008 / TC-PRODUCT-DETAIL-009
- **Environment:** Chrome, macOS, <http://localhost:5173/product/1>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Open the Product Detail page.
 2. Enter an invalid quantity, e.g., 0, -1, 1.5, e, ,, . or similar values; or leave the quantity empty.
 3. Click “Add to Cart”.
- **Expected Result:** System rejects quantity ≤ 0 , non-numeric values, empty quantity, and displays appropriate error.
- **Actual Result:** Invalid quantity is still accepted and product is added to cart instead of being rejected.

BUG-06-04: [BUG][Product Detail] - Un-auth user can add to cart

- **Issue ID:** #65
- **Severity / Priority:** Major / P1
- **Test Case:** TC-PRODUCT-DETAIL-012
- **Environment:** Chrome, macOS, <http://localhost:5173/product/1>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Open the Product Detail page when not logged in.
 2. Enter a valid quantity.
 3. Click “Add to Cart”.
- **Expected Result:** Must prompt login or redirect to the login page.
- **Actual Result:** Guest user can still successfully add the product to the cart.

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[BUG][Product Detail] - Un-auth user can add to cart #65

New issue

Open

yuran1811 opened 5 hours ago

Last edited by yuran1811

Owner

Found by Test Case

TC-PRODUCT-DETAIL-012

Requirement liên quan

FR-06, FR-08

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:5173/product/1>, [85af3ba](#)

Steps to reproduce

1. Mở trang Product Detail khi chưa đăng nhập.

2. Nhập quantity hợp lệ.

3. Bấm "Thêm vào giỏ hàng".

Expected result

Phải yêu cầu đăng nhập hoặc chuyển hướng sang màn hình đăng nhập.

Actual result

Khách văng lại vẫn thêm sản phẩm vào giỏ thành công.

Evidence

Screen.Recording.2026-06-28.at.17.16.37.mov

EShop

Giỏ hàng Đăng nhập Đăng ký

iPhone 15 Pro Max

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Điện thoại cao cấp của Apple

iPhone 15

Assignees

No one - [Assign yourself](#)

Labels

module:product

priority:P1

severity:major

status:new

type:bug

Projects

eshop-sut

Status To triage

Milestone

hww2

Due by June 29, 2026

Relationships

None yet

Development

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Evidence:

BUG-06-05: [BUG][Product Detail] - Missing breadcrumb

- **Issue ID:** #66
- **Severity / Priority:** Minor / P2
- **Test Case:** TC-PRODUCT-DETAIL-013
- **Environment:** Chrome, macOS, [http://localhost:5173/product/1, 85af3ba875c88283615e22cb108f13e2fccaf0e9](http://localhost:5173/product/1,85af3ba875c88283615e22cb108f13e2fccaf0e9)
- **Steps to Reproduce:**
 1. Open /product/:id page.
 2. Observe the navigation area at the top.
- **Expected Result:** Display full breadcrumb.
- **Actual Result:** No breadcrumb is present on the page.

yuran1811

hcmus-sw-testing--eshop-sut

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Issues 100

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[BUG][Product Detail] - Missing breadcrumb #66

New issue

Open

yuran1811 opened 5 hours ago

Last edited by yuran1811

Owner

Found by Test Case

TC-PRODUCT-DETAIL-013

Requirement liên quan

FR-23

Severity / Priority

Minor / P2

Environment

Chrome, macOS, http://localhost:5173/product/1_85af3ba

Steps to reproduce

1. Mở trang /product/:id.

2. Quan sát khu vực điều hướng phía trên.

Expected result

Hiển thị breadcrumb đầy đủ.

Actual result

Không có breadcrumb nào trên trang.

Evidence

EShop

Giỏ hàng Đăng nhập Đăng ký

iPhone 15 Pro Max

30,000,000 đ

Điện thoại cao cấp của Apple

iPhone 15

Số lượng: 1

Thêm vào giỏ hàng

Assignees

No one - [Assign yourself](#)

Labels

module:product

priority:P2

severity:minor

status:new

type:bug

Projects

eshop-sut

Status To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Notifications

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Participants

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Evidence:

FR-09

Domain Testing

Domain Testing Process:

- 1. **Identify input variables:** For the coupon application feature, the key input variables directly affecting business logic are:
 - code: Coupon code entered by the user.
 - total_amount: Current total order amount.
 - Authentication state (JWT Token): Indicates whether the user is logged in.
 - usage_count: Number of times the current user has used this coupon.
 - discount_type: Coupon type (percentage or fixed amount discount).
- 2. **Determine domain rules and Equivalence Partitioning:**
 - Partition equivalence classes for each variable (as detailed in the analysis tables below).
- 3. **Select representative values:** Select specific coupons in the system such as SAVE10 (percentage type, min order 300k, max 1 use), BIGBUY (fixed type, min order 500k, max 1 use), and VIP100 (fixed type, min order 300k, max 2 uses) as test data.
- 4. **Establish test cases:** Use the Single Fault Assumption to combine values, designing a total of 11 test cases from TC-COUPON-001 to TC-COUPON-011.

Input Domain Analysis:

Variable 1: code (Entered coupon code)

Partition Code	Partition Type	Partition Description	Representative Value
EP-CODE-1	Valid	Coupon exists in DB and is active	SAVE10
EP-CODE-2	Invalid	Coupon does not exist in DB	NOTEXIST
EP-CODE-3	Invalid	Coupon exists but is deactivated (is_active=0)	INACTIVE
EP-CODE-4	Invalid	Coupon exists but has expired	EXPIRED

Variable 2: total_amount (Total order amount)

Partition Code	Partition Type	Partition Description	Representative Value
EP-TOTAL-1	Valid	Greater than the minimum order threshold of the coupon	400,000 (SAVE10)
EP-TOTAL-2	Valid	Exactly equal to the minimum order threshold of the coupon	500,000 (BIGBUY)
EP-TOTAL-3	Invalid	Less than the minimum order threshold of the coupon	299,999 (SAVE10)
EP-TOTAL-4	Invalid	Empty order (equal to 0)	0

Variable 3: Authentication state (JWT Token)

Partition Code	Partition Type	Partition Description	Representative Value
EP-AUTH-1	Valid	User is logged in (has a valid JWT token)	Logged into test account
EP-AUTH-2	Invalid	User is not logged in (guest user)	No JWT token sent

Variable 4: User usage count (usage_count vs max_uses_per_user)

Partition Code	Partition Type	Partition Description	Representative Value
EP-USAGE-1	Valid	Usage count < maximum coupon limit	0 (SAVE10, max = 1)
EP-USAGE-2	Invalid	Usage count >= maximum coupon limit	1 (SAVE10, max = 1)

Variable 5: Coupon type (discount_type)

Partition Code	Partition Type	Partition Description	Representative Value
EP-TYPE-1	Valid	Discount by percentage (percent)	percent (SAVE10)
EP-TYPE-2	Valid	Discount by fixed amount (fixed)	fixed (BIGBUY)

List of Test Cases Designed by Domain Testing:

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
TC-COUPO N-001	Apply coupon SAVE10 (10% discount) to order of 400,000 ₪	EP-CODE-1, EP-TOTAL-1, EP-AUTH-1, EP-USAGE-1, EP-TYPE-1	Successfully applied, discount 40,000 ₪, total payment 360,000 ₪
TC-COUPO N-002	Apply coupon BIGBUY (50k discount) to order of 500,000 ₪	EP-TOTAL-2, EP-TYPE-2, other valid partitions	Successfully applied, discount 50,000 ₪, total payment 450,000 ₪
TC-COUPO N-003	Enter non-existent coupon in DB ("NOTEXIST")	EP-CODE-2	Error: "Coupon code does not exist..."
TC-COUPO N-004	Enter deactivated coupon (is_active=0)	EP-CODE-3	Error: "Coupon code does not exist..." or invalid
TC-COUPO N-005	Enter expired coupon	EP-CODE-4	Error: "Coupon has expired"
TC-COUPO N-006	Apply coupon SAVE10 to order of 299,999 ₪ (min threshold 300k)	EP-TOTAL-3	Error: "Order does not meet minimum threshold"
TC-COUPO N-007	Apply coupon SAVE10 to order of 0 ₪	EP-TOTAL-4	Error: "Order does not meet minimum threshold"
TC-COUPO N-008	Apply coupon when not logged in	EP-AUTH-2	Error: "Authentication required to apply coupon" (401 Unauthorized)
	Apply coupon SAVE10 when usage limit reached (max=1, usage=1)	EP-USAGE-2	Error: "You have already used this coupon code"

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
TC-COUPO N-009			
TC-COUPO N-010	Apply coupon VIP100 for the second time (max=2, usage=1)	EP-USAGE-1 (below limit)	Successfully applied, discount 100,000 đ
TC-COUPO N-011	Apply coupon VIP100 for the third time (max=2, usage=2)	EP-USAGE-2 (limit reached)	Error: “You have already used this coupon code”

Boundary Value Analysis (BVA)

Boundary Value Analysis (BVA) Process:

- 1. Identify boundary variables:** Variables with specific business boundaries include:
 - `total_amount`: Has a minimum boundary (`min_order_amount`) depending on each coupon.
 - `usage_count`: Has a maximum boundary (`max_uses_per_user`) depending on each coupon.
- 2. Identify boundary test points (3-value BVA or customized 2-value):**
 - **Boundary B1: total_amount for coupon SAVE10 (min = 300,000 đ):**
 - OFF (min - 1) = 299,999 đ (Rejected).
 - ON (min) = 300,000 đ (Accepted).
 - IN (min + 1) = 300,001 đ (Accepted).
 - **Boundary B2: total_amount for coupon BIGBUY (min = 500,000 đ):**
 - OFF (min - 1) = 499,999 đ (Rejected).
 - ON (min) = 500,000 đ (Accepted).
 - IN (min + 1) = 500,001 đ (Accepted).
 - **Boundary B3: usage_count for coupon SAVE10 (max = 1):**
 - ON (max - 1) = 0 uses (Accepted).
 - OFF (max) = 1 uses (Rejected).

- **Boundary B4: usage_count for coupon VIP100 (max = 2):**

- ON (max - 1) = 1 uses (Accepted).
- OFF (max) = 2 uses (Rejected).

3. Map and supplement test cases:

- The OFF boundary points are already tested in the domain testing section (TC-COUPON-006 and TC-COUPON-009).
- Link the ON/OFF/IN points with the existing test case list or add new cases (from TC-COUPON-012 to TC-COUPON-015) for full coverage.

Boundary Value Analysis Table:

Boundary B1 & B2: total_amount

- **SAVE10** (Closed minimum boundary, min = 300,000 đ, step size = 1 đ): | Boundary Test Point | Test Value | Expected Result | Linked Test Case | | :----- | :---- | :----- | :----- | | OFF (min - 1) | 299,999 đ | Rejected | TC-COUPON-006 | | ON (min) | 300,000 đ | Accepted | TC-COUPON-012 | | IN (min + 1) | 300,001 đ | Accepted | TC-COUPON-013 |
- **BIGBUY** (Closed minimum boundary, min = 500,000 đ, step size = 1 đ): | Boundary Test Point | Test Value | Expected Result | Linked Test Case | | :----- | :---- | :----- | :----- | | OFF (min - 1) | 499,999 đ | Rejected | TC-COUPON-014 | | ON (min) | 500,000 đ | Accepted | TC-COUPON-002 | | IN (min + 1) | 500,001 đ | Accepted | TC-COUPON-015 |

Boundary B3 & B4: usage_count (Usage count)

- **SAVE10** (Closed maximum boundary, max = 1, step size = 1): | Boundary Test Point | Test Value | Expected Result | Linked Test Case | | :----- | :---- | :----- | :----- | | ON (max - 1) | 0 uses | Accepted | TC-COUPON-001 | | OFF (max) | 1 uses | Rejected | TC-COUPON-009 |
- **VIP100** (Closed maximum boundary, max = 2, step size = 1): | Boundary Test Point | Test Value | Expected Result | Linked Test Case | | :----- | :---- | :----- | :----- | | ON (max - 1) | 1 uses | Accepted | TC-COUPON-010 | | OFF (max) | 2 uses | Rejected | TC-COUPON-011 |

List of Test Cases Designed by BVA:

Test Case ID	Test Case Description	Tested Boundary	Boundary Point	Value	Expected Result
TC-COUPON-006	Apply coupon SAVE10 to order of 299,999 ₪	B1 (SAVE10 min)	OFF	299,999 ₪	Rejected
TC-COUPON-012	Apply coupon SAVE10 to order of 300,000 ₪	B1 (SAVE10 min)	ON	300,000 ₪	Accepted and discount 30k
TC-COUPON-013	Apply coupon SAVE10 to order of 300,001 ₪	B1 (SAVE10 min)	IN	300,001 ₪	Accepted and discount 30k
TC-COUPON-014	Apply coupon BIGBUY to order of 499,999 ₪	B2 (BIGBUY min)	OFF	499,999 ₪	Rejected
TC-COUPON-002	Apply coupon BIGBUY to order of 500,000 ₪	B2 (BIGBUY min)	ON	500,000 ₪	Accepted and discount 50k
TC-COUPON-015	Apply coupon BIGBUY to order of 500,001 ₪	B2 (BIGBUY min)	IN	500,001 ₪	Accepted and discount 50k
TC-COUPON-001	Apply coupon SAVE10 when usage_count = 0	B3 (SAVE10 max)	ON	0 uses	Accepted
TC-COUPON-009	Apply coupon SAVE10 when usage_count = 1	B3 (SAVE10 max)	OFF	1 uses	Rejected
TC-COUPON-010	Apply coupon VIP100 when usage_count = 1	B4 (VIP100 max)	ON	1 uses	Accepted
TC-COUPON-011	Apply coupon VIP100 when usage_count = 2	B4 (VIP100 max)	OFF	2 uses	Rejected

AI Gap Analysis

For the Coupon Application feature (FR-09), the AI exhibited significant gaps in logical reasoning and boundary testing:

- **Violation of the Single Fault Assumption:** The test case TC-COUPON-011 initially designed by the AI combined two separate faults (an expired coupon and an order amount below the minimum threshold). This makes debugging and isolating the root cause ambiguous when the test case fails.
- **Discrepancy in representative values:** In the partition for order amount below the minimum threshold (EP-TOTAL-3), the representative value was declared as 299,999 ₺, but in the actual test case TC-COUPON-006, the AI used a value of 200,000 ₺, breaking traceability.
- **Incorrect selection of valid partition test points:** For the valid order amount partition EP-TOTAL-1, the AI selected 300,000 ₺ as the representative value. This is actually the boundary ON point for the SAVE10 coupon rather than an interior point (IN point) deep within the partition. This causes unnecessary overlap with BVA.
- **Redundant cases and missing boundaries:** The AI generated redundant test cases (e.g., TC-COUPON-013 and TC-COUPON-017, which did not provide any new coverage) while completely omitting boundary testing for the BIGBUY coupon (500,000 ₺ threshold), empty coupon code input (" "), and the maximum usage boundary for the VIP100 coupon (max = 2).

Root cause: When facing features with multiple co-dependent constraints (such as active duration, minimum order threshold, and user usage limits), the AI tends to suffer from context overload, leading it to combine multiple faults in a single test case, select representative values that overlap with boundary points, and fail to optimize the total number of test cases.

Bug Report

Bug ID	Title	Severity	Detecting Test Case	GitHub Link
BUG-09-01	[BUG][Coupon] - SAVE10 make the final price too high (wrong calculation)	Major / P1	TC-COUPON-001, TC-COUPON-013	Issue #67
BUG-09-02	[BUG][Coupon] - BIGBUY be rejected at its threshold	Major / P1	TC-COUPON-002, TC-COUPON-012	Issue #69
	[BUG][Coupon] - Un-auth user can apply the coupon	Major / P1	TC-COUPON-008	Issue #97

Bug ID	Title	Severity	Detecting Test Case	GitHub Link
BUG-09-03				

BUG-09-01: [BUG][Coupon] - SAVE10 make the final price too high (wrong calculation)

- **Issue ID:** #67
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-001, TC-COUPON-013
- **Environment:** Chrome, macOS, <http://localhost:5173/checkout>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Apply coupon SAVE10 to a valid order.
 2. Observe the discount amount and total payment.
- **Expected Result:** Percentage discount must be calculated correctly based on discount_value.
- **Actual Result:** Backend uses an incorrect formula that causes the total invoice amount to spike.

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[BUG][Coupon] - SAVE10 make the final price too high (wrong calculation) #67

New issue

Open

yuran1811 opened 4 hours ago

Last edited by yuran1811

Owner

Found by Test Case

TC-COUPON-001, TC-COUPON-013

Requirement liên quan

FR-09

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:5173/checkout>, 85af3ba

Steps to reproduce

1. Áp dụng mã SAVE10 cho đơn hàng hợp lệ.

2. Quan sát số tiền giảm và tổng thanh toán.

Expected result

Giảm giá phần trăm phải tính đúng theo discount_value.

Actual result

Backend dùng công thức sai và làm tổng hóa đơn tăng vọt.

Evidence

Xác Nhận Đơn Hàng

Sản phẩm:

- iPhone 15 Pro Max x 1 — 30,000,000 đ

Tổng tiền thanh toán (VND):

30000000

Mã Giảm Giá

SAVE10

Áp dụng

✓ Áp dụng thành công! Giảm 10%

Tiết kiệm: -270,000,000 đ

Thành tiền: 300,000,000 đ

Assignees

No one - [Assign yourself](#)

Labels

module:coupon

priority:P1

severity:major

status:new

type:bug

Projects

eshop-sut

Status

To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Delete issue

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Evidence:

BUG-09-02: [BUG][Coupon] - BIGBUY be rejected at its threshold

- **Issue ID:** #69
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-002, TC-COUPON-012
- **Environment:** Chrome, macOS, <http://localhost:5173/checkout>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Apply coupon BIGBUY to an order exactly equal to min_order_amount.
 2. Check the system response.
- **Expected Result:** Orders matching the exact minimum threshold must be accepted.
- **Actual Result:** The system rejects the coupon due to using a > comparison instead of >=.

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[BUG][Coupon] - BIGBUY be rejected at its threshold #69

New issue

Open

yuran1811 opened 4 hours ago

Last edited by yuran1811

Owner

Found by Test Case

TC-COUPON-002, TC-COUPON-012

Requirement liên quan

FR-09

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:5173/checkout>, [85af3ba](#)

Steps to reproduce

1. Áp dụng mã BIGBUY cho đơn hàng đúng min_order_amount.

2. Kiểm tra phản hồi từ hệ thống.

Expected result

Đơn hàng bằng đúng ngưỡng tối thiểu phải được chấp nhận.

Actual result

Hệ thống từ chối do so sánh > thay vì >= .

Evidence

Xác Nhận Đơn Hàng

Sản phẩm:

- 500k x 1 — 500,000 đ

Tổng tiền thanh toán (VND):

500000

Mã Giảm Giá

BIGBUY

Áp dụng

Đơn hàng chưa đủ giá trị tối thiểu 500,000 đ để áp dụng mã này

Tổng thanh toán: 500,000 đ

Assignees

No one - [Assign yourself](#)

Labels

module:coupon

priority:P1

severity:major

status:new

type:bug

Projects

eshop-sut

Status

To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

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Clone issue

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Delete issue

Give feedback

Evidence:

BUG-09-03: [BUG][Coupon] - Un-auth user can apply the coupon

- **Issue ID:** #97
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-008
- **Environment:** Chrome, macOS, <http://localhost:5173/checkout>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Call API /api/apply-coupon when not logged in.
 2. Observe the system response.
- **Expected Result:** Must return an authentication error.
- **Actual Result:** API allows unauthenticated guest users to successfully apply the coupon.

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[BUG][Coupon] - Un-auth user can apply the coupon #97

New issue

Open

yuran1811 opened 2 hours ago

Owner

Found by Test Case

TC-COUPON-008

Requirement liên quan

FR-09

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:5173/checkout>, [85af3ba](#)

Steps to reproduce

1. Gọi API /api/apply-coupon khi chưa đăng nhập.

2. Quan sát phản hồi hệ thống.

Expected result

Phải trả về lỗi xác thực.

Actual result

API cho phép khách chưa đăng nhập áp dụng coupon thành công.

Evidence

EShop

Giỏ hàng

Đăng nhập

Đăng ký

Xác Nhận Đơn Hàng

Sản phẩm:

Tổng tiền thanh toán (VND):

500001

Mã Giảm Giá

BIGBUY

Áp dụng

✓ Áp dụng thành công! Giảm 50,000 đ

Tiết kiệm: 50,000 đ

Thành tiền: 450,001 đ

Assignees

No one - [Assign yourself](#)

Labels

module:coupon

priority:P2

severity:minor

status:new

type:bug

Projects

eshop-sut

Status

To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Give feedback

Evidence:

FR-17

Domain Testing

Domain Testing Process:

1. Identify input variables for coupon creation/management (Admin):

- code: Coupon code (string).
- type: Coupon type (percent or fixed).
- discount_value: Discount value (float/integer).
- expired_at: Expiration date (date string).
- min_order_amount: Minimum order amount (float/integer).
- max_uses_per_user: Maximum uses per user (integer).
- Authentication / Authorization: Admin API access permission (JWT token: Admin, User, Guest).

2. Determine domain rules and Equivalence Partitioning:

- Partition valid and invalid equivalence classes for each variable (as detailed in the tables below).

3. Select representative values: Select specific values within each partition to design test cases.

4. Establish test cases: Apply the Single Fault Assumption, designing 17 primary test cases (TC-COUPON-ADMIN-001 to TC-COUPON-ADMIN-017) and additional cases covering boundaries, formats, and security.

Input Domain Analysis:

Variable 1: code (Coupon code)

Partition Code	Partition Type	Partition Description	Representative Value
EP-CODE-1	Valid	Unique, non-empty string	" SUMMER25 "
EP-CODE-2	Invalid	Empty string (field left blank)	" "
EP-CODE-3	Invalid	Duplicates an existing coupon code in the DB	" SAVE10 "

Variable 2: type (Discount type)

Partition Code	Partition Type	Partition Description	Representative Value
EP-TYPE-1	Valid	Percentage type (percent)	"percent"
EP-TYPE-2	Valid	Fixed amount type (fixed)	"fixed"
EP-TYPE-3	Invalid	Value outside the valid set {percent, fixed}	"voucher"

Variable 3: discount_value (Discount value)

Partition Code	Partition Type	Partition Description	Representative Value
EP-DV-1	Valid	Positive number (> 0)	15 (for percent) or 50000 (for fixed)
EP-DV-2	Invalid	Value is 0	0
EP-DV-3	Invalid	Negative value	-10

Variable 4: expired_at (Expiration date)

Partition Code	Partition Type	Partition Description	Representative Value
EP-EXP-1	Valid	Date in the future (ISO format YYYY-MM-DD)	"2099-12-31"
EP-EXP-2	Invalid	Required field left blank	" "
EP-EXP-3	Invalid	Incorrect date format	"31-12-2099"

Variable 5: min_order_amount (Minimum order threshold)

Partition Code	Partition Type	Partition Description	Representative Value
EP-MOA-1	Valid	Equal to 0 (no minimum order threshold required)	0
EP-MOA-2	Valid	Positive number (> 0)	200,000
EP-MOA-3	Invalid	Negative number	-1

Variable 6: max_uses_per_user (Maximum uses per user)

Partition Code	Partition Type	Partition Description	Representative Value
EP-MPU-1	Valid	Equal to 1	1
EP-MPU-2	Valid	Positive integer > 1	5
EP-MPU-3	Invalid	Equal to 0	0
EP-MPU-4	Invalid	Negative number	- 1

Variable 7: Authentication / Authorization

Partition Code	Partition Type	Partition Description	Representative Value
EP-ROLE-1	Valid	Logged in with Admin role	admin@eshop.com
EP-ROLE-2	Invalid	Not logged in (no JWT Token sent)	No token sent
EP-ROLE-3	Invalid	Logged in with User role (no admin permissions)	test@eshop.com

List of Test Cases Designed by Domain Testing:

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
TC-COUPON-ADMIN-001	Successfully create a percentage coupon	All valid EP variables (EP-CODE-1, EP-TYPE-1, EP-DV-1, EP-EXP-1, EP-MOA-2, EP-MPU-2, EP-ROLE-1)	Return status 201/200; coupon saved in DB
TC-COUPON-ADMIN-002	Successfully create a fixed coupon	EP-TYPE-2, other valid EP variables	Return status 201/200; coupon saved in DB
TC-COUPON		EP-CODE-2	

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
-ADMIN-003	Leave coupon code blank when creating coupon		Return status 400 Bad Request and error code empty
TC-COUPON-ADMIN-004	Create coupon with duplicate code (SAVE10)	EP-CODE-3	Return status 400 Bad Request and error code duplicates
TC-COUPON-ADMIN-005	Create coupon with invalid type ("voucher")	EP-TYPE-3	Return status 400 Bad Request and error invalid type
TC-COUPON-ADMIN-006	Create coupon with discount_value = 0	EP-DV-2	Return status 400 and error discount must be positive
TC-COUPON-ADMIN-007	Create coupon with negative discount_value (-10)	EP-DV-3	Return status 400 and error discount must be positive
TC-COUPON-ADMIN-008	Leave expired_at blank when creating coupon	EP-EXP-2	Return status 400 and error expiration date required
TC-COUPON-ADMIN-009	Create coupon with invalid expired_at format ("31-12-2099")	EP-EXP-3	Return status 400 and error invalid date format
TC-COUPON	Create coupon with negative min_order_amount (-1)	EP-MOA-3	Return status 400 and error minimum threshold cannot be negative

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
-ADMIN-010			
TC-COUPON-ADMIN-011	Create coupon with max_uses_per_user = 0	EP-MPU-3	Return status 400 and error max uses must be >= 1
TC-COUPON-ADMIN-012	Create coupon with negative max_uses_per_user (-1)	EP-MPU-4	Return status 400 and error max uses must be >= 1
TC-COUPON-ADMIN-013	Delete coupon with a valid and existing ID	Delete operation, EP-ROLE-1	Return status 200/204; coupon deleted from system
TC-COUPON-ADMIN-014	Delete coupon with a non-existent ID	Delete operation, EP-ROLE-1	Return status 404 Not Found
TC-COUPON-ADMIN-015	Get list of coupons	Read operation, EP-ROLE-1	Return status 200 and list of all coupons
TC-COUPON-ADMIN-016	Create coupon when not logged in	EP-ROLE-2	Return status 401 Unauthorized
TC-COUPON	Create coupon with user role	EP-ROLE-3	Return status 403 Forbidden

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
-ADMIN-017			
TC-COUPON-ADMIN-024	Create coupon with expired_at in the past	Exploratory test on expiration date	Return status 400 and reject past date
TC-COUPON-ADMIN-025	POST payload missing code field completely in body	Missing required field	Return status 400 and error missing required field
TC-COUPON-ADMIN-026	Create coupon with code containing malicious script (XSS testing)	Security testing (XSS)	System stores or displays safely escaped value without rendering HTML

Boundary Value Analysis (BVA)

Boundary Value Analysis (BVA) Process:

- 1. Identify boundary variables:** Numeric input fields with a minimum boundary include:
 - discount_value: Must be greater than 0 (> 0). Open minimum boundary.
 - min_order_amount: Must be non-negative (>= 0). Closed minimum boundary.
 - max_uses_per_user: Must be greater than or equal to 1 (>= 1). Closed minimum boundary.
- 2. Identify boundary test points (3-value BVA):**
 - **Boundary B1: discount_value (open boundary > 0, step size = 1):**
 - OFF (boundary) = 0 (Rejected).
 - ON (min + 1) = 1 (Accepted).
 - IN (min + 2) = 2 (Accepted).
 - **Boundary B2: min_order_amount (closed boundary >= 0, step size = 1):**
 - OFF (min - 1) = -1 (Rejected).

- ON (min) = 0 (Accepted).
- IN (min + 1) = 1 (Accepted).
- **Boundary B3: max_uses_per_user (closed boundary >= 1, step size = 1):**
 - OFF (min - 1) = 0 (Rejected).
 - ON (min) = 1 (Accepted).
 - IN (min + 1) = 2 (Accepted).

3. Map and supplement test cases:

- Link boundary OFF points already tested in domain testing.
- Add ON and IN test cases (from TC-COUPON-ADMIN-018 to TC-COUPON-ADMIN-023) to fully verify boundary behaviors.

Boundary Value Analysis Table:

Boundary B1: discount_value (open boundary > 0)

Boundary Test Point	Test Value	Expected Result	Linked Test Case
OFF (boundary)	0	Rejected	TC-COUPON-ADMIN-006
ON (min)	1	Accepted and successfully created	TC-COUPON-ADMIN-018
IN (min + 1)	2	Accepted and successfully created	TC-COUPON-ADMIN-019

Boundary B2: min_order_amount (closed boundary >= 0)

Boundary Test Point	Test Value	Expected Result	Linked Test Case
OFF (min - 1)	- 1	Rejected	TC-COUPON-ADMIN-010
ON (min)	0	Accepted and successfully created	TC-COUPON-ADMIN-020
IN (min + 1)	1	Accepted and successfully created	TC-COUPON-ADMIN-021

Boundary B3: max_uses_per_user (closed boundary >= 1)

Boundary Test Point	Test Value	Expected Result	Linked Test Case
OFF (min - 1)	0	Rejected	TC-COUPON-ADMIN-011

Boundary Test Point	Test Value	Expected Result	Linked Test Case
ON (min)	1	Accepted and successfully created	TC-COUPON-ADMIN-022
IN (min + 1)	2	Accepted and successfully created	TC-COUPON-ADMIN-023

List of Test Cases Designed by BVA:

Test Case ID	Test Case Description	Tested Boundary	Boundary Point	Value	Expected Result
TC-COUPON-ADMIN-006	Create coupon with discount_value = 0	B1	OFF	0	Rejected
TC-COUPON-ADMIN-018	Create coupon with discount_value = 1 (lowest positive value)	B1	ON	1	Accepted and successfully created
TC-COUPON-ADMIN-019	Create coupon with discount_value = 2 (just above boundary)	B1	IN	2	Accepted and successfully created
TC-COUPON-ADMIN-010	Create coupon with min_order_amount = -1	B2	OFF	-1	Rejected
TC-COUPON-ADMIN-020	Create coupon with min_order_amount = 0	B2	ON	0	Accepted and successfully created
TC-COUPON-ADMIN-021	Create coupon with min_order_amount = 1	B2	IN	1	Accepted and successfully created
TC-COUPON-ADMIN-011	Create coupon with max_uses_per_user = 0	B3	OFF	0	Rejected
TC-COUPON-ADMIN-022	Create coupon with max_uses_per_user = 1	B3	ON	1	Accepted and successfully created
TC-COUPON-ADMIN-023	Create coupon with max_uses_per_user = 2	B3	IN	2	Accepted and successfully created

AI Gap Analysis

The Admin Coupon Management feature (FR-17) carries high complexity regarding roles, authorization, and input validation. The AI left behind several gaps in testing:

- **Severe redundancies between DT and BVA:** The AI designed 3 completely redundant test case pairs where the BVA OFF cases duplicated DT Invalid cases. These are: TC-COUPON-ADMIN-018 duplicated TC-COUPON-ADMIN-006 (discount = 0), TC-COUPON-ADMIN-021 duplicated TC-COUPON-ADMIN-010 (min_order < 0), and TC-COUPON-ADMIN-024 duplicated TC-COUPON-ADMIN-011 (max_uses = 0) in the original run.
- **Endpoint Hallucination:** The AI assumed the coupon listing endpoint was GET /api/admin/coupons. However, according to the actual API specification, the correct endpoint is GET /api/coupons, which handles both client and admin requests based on authorization middleware.
- **Omissions of real-world scenarios:** The AI missed critical test cases such as creating coupons with an expiration date in the past, sending a POST body completely missing the code property (rather than just sending " "), inputting non-integer values for maximum uses, and detailed validation of all fields in the response body.

Root cause: The AI relied on generic RESTful API conventions (assuming admin routes are prefixed with /api/admin/...), hallucinating endpoints instead of strictly verifying the provided API specifications. It also failed to deduplicate overlapping test cases between Domain Testing and Boundary Value Analysis.

Bug Report

Bug ID	Title	Severity	Detecting Test Case	GitHub Link
BUG-17-01	[BUG][Coupon Admin] - Allow to create coupon with type="invalid"	Major / P1	TC-COUPON-ADMIN-005	Issue #98
BUG-17-02	[BUG][Coupon Admin] - Allow to create coupon with discount_value=0	Major / P1	TC-COUPON-ADMIN-006	Issue #99
BUG-17-03	[BUG][Coupon Admin] - Allow to create coupon with discount_value<0	Major / P1	TC-COUPON-ADMIN-007	Issue #100

Bug ID	Title	Severity	Detecting Test Case	GitHub Link
BUG-17-04	[BUG][Coupon Admin] - Allow to create coupon with invalid expired_at	Major / P1	TC-COUPON-ADMIN-009	Issue #101
BUG-17-05	[BUG][Coupon Admin] - Allow to create coupon with min_order_amount<0	Major / P1	TC-COUPON-ADMIN-010	Issue #102
BUG-17-06	[BUG][Coupon Admin] - Allow to create coupon with max_uses_per_user=0	Major / P1	TC-COUPON-ADMIN-011	Issue #103
BUG-17-07	[BUG][Coupon Admin] - Allow to create coupon with max_uses_per_user<0	Major / P1	TC-COUPON-ADMIN-012	Issue #104
BUG-17-08	[BUG][Coupon Admin] - Delete non-exist coupon still get status 200	Major / P1	TC-COUPON-ADMIN-014	Issue #105
BUG-17-09	[BUG][Coupon Admin] - Non-admin user can create coupon	Major / P1	TC-COUPON-ADMIN-017	Issue #106
BUG-17-10	[BUG][Coupon Admin] - Allow to create coupon without code field	Major / P1	TC-COUPON-ADMIN-025	Issue #109

BUG-17-01: [BUG][Coupon Admin] - Allow to create coupon with type="invalid"

- **Issue ID:** #98
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-005

- **Environment:** Chrome, macOS, <http://localhost:5174>, 85af3ba875c88283615e22cb108f13e2fccaf0e9

- **Steps to Reproduce:**

1. Send POST /api/admin/coupons with type = voucher.
2. Check the status code.

- **Expected Result:** Backend must reject the invalid type.

- **Actual Result:** Backend accepts the type “voucher” and returns 200.

BUG-17-02: [BUG][Coupon Admin] - Allow to create coupon with discount_value=0

- **Issue ID:** #99
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-006
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send POST /api/admin/coupons with discount_value = 0.
 2. Check the status code.
- **Expected Result:** Backend must reject discount_value = 0.
- **Actual Result:** Backend accepts the request and returns 200.

BUG-17-03: [BUG][Coupon Admin] - Allow to create coupon with discount_value<0

- **Issue ID:** #100
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-007
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send POST /api/admin/coupons with a negative discount_value.
 2. Check the status code.
- **Expected Result:** Backend must reject negative values.
- **Actual Result:** Backend accepts negative discount_value and returns 200.

BUG-17-04: [BUG][Coupon Admin] - Allow to create coupon with invalid expired_at

- **Issue ID:** #101
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-009
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send POST /api/admin/coupons with expired_at = 31-12-2099.
 2. Check the status code.
- **Expected Result:** Backend must reject the invalid date format.
- **Actual Result:** Backend accepts the invalid date format and returns 200.

BUG-17-05: [BUG][Coupon Admin] - Allow to create coupon with min_order_amount<0

- **Issue ID:** #102
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-010
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send POST /api/admin/coupons with min_order_amount = -1.
 2. Check the status code.
- **Expected Result:** Backend must reject negative values.
- **Actual Result:** Backend accepts negative min_order_amount and returns 200.

BUG-17-06: [BUG][Coupon Admin] - Allow to create coupon with max_uses_per_user=0

- **Issue ID:** #103
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-011
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send POST /api/admin/coupons with max_uses_per_user = 0.
 2. Check the status code.
- **Expected Result:** Backend must reject the value 0.
- **Actual Result:** Backend accepts max_uses_per_user = 0 and returns 200.

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[BUG][Coupon Admin] - Allow to create coupon with max_uses_per_user=0 #103

New issue

Open

yuran1811 opened 1 hour ago

Last edited by yuran1811

Owner

Found by Test Case

TC-COUPON-ADMIN-011

Requirement liên quan

FR-17

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:5174>, [85af3ba](#)

Steps to reproduce

1. Gửi POST /api/admin/coupons với max_uses_per_user = 0.

2. Kiểm tra status code.

Expected result

Backend phải từ chối giá trị 0.

Actual result

Backend chấp nhận max_uses_per_user = 0 và trả về 200.

Evidence

Send Request

POST <http://localhost:3000/api/admin/coupons>

Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZiI6Im5wcm9sZSI6ImFkbWUiOiwiWF0iOiJoxNzgyNjQxMjIyYQ.ZwtSSyY6vJGzAsobSX0xq0L15Mvp0kP2j0v70n8IJM

Content-Type: application/json

1 HTTP/1.1 200 OK

2 X-Powered-By: Express

3 Access-Control-Allow-Origin: *

4 Content-Type: application/json; charset=utf-8

5 Content-Length: 36

6 ETag: W/"24-fQM06ETbaHU60MSxJeRTVEyal0o"

7 Date: Sun, 28 Jun 2026 14:50:48 GMT

8 Connection: close

9

10 {

11 "message": "Coupon created",

12 "id": 12

13 }

"code": "INVALIDVALS",

"type": "invalid",

"discount_value": -10000,

"min_order_amount": -200000,

"expired_at": "31-12-2099",

"max_uses_per_user": 0

Create sub-issue

Assignees

No one - [Assign yourself](#)

Labels

module:admin

priority:P1

severity:major

status:new

type:bug

Projects

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Status

To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Evidence:

BUG-17-07: [BUG][Coupon Admin] - Allow to create coupon with max_uses_per_user<0

- **Issue ID:** #104
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-012
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send POST /api/admin/coupons with a negative max_uses_per_user.
 2. Check the status code.
- **Expected Result:** Backend must reject negative values.
- **Actual Result:** Backend accepts negative max_uses_per_user and returns 200.

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[BUG][Coupon Admin] - Allow to create coupon with max_uses_per_user<0 #104

New issue

Open

yuran1811 opened 1 hour ago

Last edited by yuran1811

Owner

Found by Test Case

TC-COUPON-ADMIN-012

Requirement liên quan

FR-17

Severity / Priority

Major / P1

Environment

Chrome, macOS, http://localhost:5174, 85af3ba

Steps to reproduce

1. Gửi POST /api/admin/coupons với max_uses_per_user âm.
2. Kiểm tra status code.

Expected result

Backend phải từ chối giá trị âm.

Actual result

Backend chấp nhận max_uses_per_user âm và trả về 200.

Evidence

Send Request

POST http://localhost:3000/api/admin/coupons

Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZi6MSwicz9sZSI6ImFkbWl1aWwMF0IjoxMzgyNjQzMllyQjZwSS9y6vJ6zAsobSj0xq8L1SMvp8kP2j0v76n8LJMContent-Type: application/json

1 HTTP/1.1 200 OK

2 X-Powered-By: Express

3 Access-Control-Allow-Origin: *

4 Content-Type: application/json; charset=utf-8

5 Content-Length: 34

6 ETag: W/"24-Co8yu+Mh6tR/jeCxiRiLKV7gtM"

7 Date: Sun, 28 Jun 2026 14:53:46 GMT

8 Connection: close

9

10 {

11 "message": "Coupon created",

12 "id": 13

13 }

Create sub-issue

Assignees

No one - [Assign yourself](#)

Labels

module:admin priority:P1 severity:major status:new type:bug

Projects

eshop-sut

Status To triage

Milestone

hww2

Due by June 29, 2026

Relationships

None yet

Development

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🗑 Delete issue

💬 Give feedback

Evidence:

BUG-17-08: [BUG][Coupon Admin] - Delete non-exist coupon still get status 200

- **Issue ID:** #105
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-014
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send DELETE /api/admin/coupons/999999.
 2. Check the status code.
- **Expected Result:** Must return 4xx (e.g., 404 Not Found) when the coupon does not exist.
- **Actual Result:** API returns 200 instead of 404/4xx.

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[BUG][Coupon Admin] - Delete non-exist coupon still get status 200 #105

New issue

Open

yuran1811 opened 1 hour ago

Owner

Found by Test Case

TC-COUPON-ADMIN-014

Requirement liên quan

FR-17

Severity / Priority

Major / P1

Environment

Chrome, macOS, http://localhost:5174, 85af3ba

Steps to reproduce

1. Gửi DELETE /api/admin/coupons/999999.
2. Kiểm tra status code.

Expected result

Phải trả về 4xx khi coupon không tồn tại.

Actual result

API trả về 200 thay vì 404/4xx.

Evidence

Send Request

DELETE http://localhost:3000/api/admin/coupons/999999
Authorization: Bearer eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJpZiZCI6Im9sZSIsImFkbWludlwiOiJ0Ij0xMzgyNjQzMllyfQ.ZwtSS9jY6vJGzAsobSX0xq8L1SMvp0kP2j0v76n8LJM

1 HTTP/1.1 200 OK
2 X-Powered-By: Express
3 Access-Control-Allow-Origin: *
4 Content-Type: application/json; charset=utf-8
5 Content-Length: 28
6 ETag: W/"1c-q1LucapgM0Q+46/o9M8fayoQTxQ"
7 Date: Sun, 28 Jun 2026 14:55:29 GMT
8 Connection: close
9
10 {
11 "message": "Coupon deleted"
12 }

Create sub-issue

yuran1811 added this to the hw2 milestone 1 hour ago

Assignees

No one - [Assign yourself](#)

Labels

module:admin

priority:P1

severity:major

status:new

type:bug

Projects

eshop-sut

Status

To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Give feedback

Evidence:

BUG-17-09: [BUG][Coupon Admin] - Non-admin user can create coupon

- **Issue ID:** #106
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-017
- **Environment:** Chrome, macOS, http://localhost:5174, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Log in with a user role account.
 2. Send POST /api/admin/coupons.
- **Expected Result:** Non-admin must receive 403 Forbidden.
- **Actual Result:** Authorization middleware is missing and request returns 200.

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[BUG][Coupon Admin] - Non-admin user can create coupon #106

New issue

🔓 Open

yuran1811 opened 1 hour ago

Owner

Found by Test Case

TC-COUPON-ADMIN-017

Requirement liên quan

FR-17

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:5174>, [85af3ba](#)

Steps to reproduce

1. Đăng nhập bằng tài khoản role user.

2. Gửi POST /api/admin/coupons.

Expected result

Non-admin phải nhận 403 Forbidden.

Actual result

Middleware phân quyền thiếu và request vẫn trả về 200.

Evidence

Send Request

GET http://localhost:3000/api/users/me

Authorization: Bearer ey.J0Gci0JiUzIiNiIsInRScCI6IkpXVCJ9.eyJpZCI6Imlicm9sZSI6InVzZXIiLCJpYXQiOiE3ODI2NTg2IjZ9.SiKqFWUB2t13UCLUkfyM1A6PwoDPWN41B7RTagJyCl

###

Send Request

POST http://localhost:3000/api/admin/coupons

Authorization: Bearer ey.J0Gci0JiUzIiNiIsInRScCI6IkpXVCJ9.eyJpZCI6Imlicm9sZSI6InVzZXIiLCJpYXQiOiE3ODI2NTg2IjZ9.SiKqFWUB2t13UCLUkfyM1A6PwoDPWN41B7RTagJyCl

Content-Type: application/json

{

"code": "USER",

"type": "Invalid",

"discount_value": 10000

}

1 HTTP/1.1 200 OK

2 X-Powered-By: Express

3 Access-Control-Allow-Origin: *

4 Content-Type: application/json; charset=utf-8

5 Content-Length: 36

6 ETag: W/"24-//b/3DeWkI4/4gzjs0/Zpft8R0o"

7 Date: Sun, 28 Jun 2026 14:59:21 GMT

8 Connection: close

9

10 {

11 "message": "Coupon created",

12 "id": 14

13 }

Assignees

No one - [Assign yourself](#)

Labels

module:admin priority:P1 severity:major status:new type:bug

Projects

eshop-sut

Status To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Evidence:

BUG-17-10: [BUG][Coupon Admin] - Allow to create coupon without code field

- **Issue ID:** #109
- **Severity / Priority:** Major / P1
- **Test Case:** TC-COUPON-ADMIN-025
- **Environment:** Chrome, macOS, <http://localhost:5174>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Send POST /api/admin/coupons without the code field.
 2. Check the status code.
- **Expected Result:** Missing required fields must be rejected.
- **Actual Result:** Backend accepts the request and returns 200.

FR-20

Domain Testing

Domain Testing Process:

- 1. Identify input variables:** For the mobile cart feature, the key input variables and interactions affecting cart logic are:
 - `add_quantity`: Quantity added to cart from the product detail page.
 - `edit_quantity`: Quantity edited directly in the quantity input field on the cart page.
 - `auth_state`: Authentication state (logged in) when the user clicks the checkout button.
- 2. Determine domain rules and Equivalence Partitioning:**
 - Partition valid and invalid equivalence classes for each variable (as detailed in the tables below).
- 3. Select representative values:** Select specific representative values (e.g., 2, 0, -1, 1.5, "", "abc") for both adding and editing quantities, and the login state (logged in / guest).
- 4. Establish test cases:** Use the Single Fault Assumption to combine values, designing a total of 17 test cases (TC-CART-MOBILE-001 to TC-CART-MOBILE-017).

Input Domain Analysis:

Variable 1: `add_quantity` (Quantity added from product detail page)

Partition Code	Partition Type	Partition Description	Representative Value
EP-AQ-1	Valid	Positive integer ≥ 1	2
EP-AQ-2	Invalid	Value is 0	0
EP-AQ-3	Invalid	Negative integer	-1
EP-AQ-4	Invalid	Decimal number	1.5
EP-AQ-5	Invalid	Input field left blank	" "
EP-AQ-6	Invalid	Non-numeric character string	"abc"

Variable 2: edit_quantity (Quantity edited directly in the cart)

Partition Code	Partition Type	Partition Description	Representative Value
EP-EQ-1	Valid	Positive integer ≥ 1	2
EP-EQ-2	Invalid	Value is 0	0
EP-EQ-3	Invalid	Negative integer	- 1
EP-EQ-4	Invalid	Decimal number	1 . 5
EP-EQ-5	Invalid	Input field left blank	" "
EP-EQ-6	Invalid	Non-numeric character string	" abc "

Variable 3: auth_state (Authentication state during checkout)

Partition Code	Partition Type	Partition Description	Representative Value
EP-AUTH-1	Valid	User is logged in (has a valid JWT token)	Logged into test account
EP-AUTH-2	Invalid	Unauthenticated guest user	Not logged in

List of Test Cases Designed by Domain Testing:

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
TC-CART-MOBILE-001	Add product from Home screen (default quantity = 1)	EP-AQ-1 (default)	Successfully add product with quantity 1
TC-CART-MOBILE-002	Add product from detail page with valid quantity (2)	EP-AQ-1	Successfully add product with quantity 2
TC-CART-MOBILE-003	Add product from detail page with quantity = 0	EP-AQ-2	Reject or automatically normalize quantity to 1
TC-CART-MOBILE-004	Add product from detail page with negative quantity (-1)	EP-AQ-3	Reject or automatically normalize quantity to 1

Test Case ID	Test Case Description	Tested Partition(s)	Expected Result
TC-CART-MOBILE-005	Add product from detail page with decimal quantity (1.5)	EP-AQ-4	Reject or automatically normalize quantity to 1
TC-CART-MOBILE-006	Add product from detail page with empty quantity	EP-AQ-5	Reject or automatically normalize quantity to 1
TC-CART-MOBILE-007	Add product from detail page with non-numeric quantity ("abc")	EP-AQ-6	Reject or automatically normalize quantity to 1
TC-CART-MOBILE-008	Edit quantity directly in cart to valid quantity (2)	EP-EQ-1	Successfully update product quantity in cart to 2
TC-CART-MOBILE-009	Edit quantity directly in cart to 0	EP-EQ-2	Reject or automatically normalize quantity to 1 (or prompt confirm deletion)
TC-CART-MOBILE-010	Edit quantity directly in cart to negative quantity (-1)	EP-EQ-3	Reject or automatically normalize quantity to 1
TC-CART-MOBILE-011	Edit quantity directly in cart to decimal quantity (1.5)	EP-EQ-4	Reject or automatically normalize to a valid integer
TC-CART-MOBILE-012	Edit quantity directly in cart to empty	EP-EQ-5	Reject or automatically normalize quantity to 1
TC-CART-MOBILE-013	Edit quantity directly in cart to non-numeric quantity ("abc")	EP-EQ-6	Reject or automatically normalize quantity to 1
TC-CART-MOBILE-014	Click "Remove" button to remove product from cart	Click remove operation	Display confirm deletion dialog; clicking yes removes item from cart
TC-CART-MOBILE-015	Click "← Continue Shopping" link	Click link operation	Redirect to Home page
TC-CART-MOBILE-016	Unauthenticated user clicks checkout button	EP-AUTH-2	Block checkout, display warning, and redirect to login page
TC-CART-MOBILE-017	Authenticated user clicks checkout button	EP-AUTH-1	Successfully redirect to Checkout page

Boundary Value Analysis (BVA)

Boundary Value Analysis (BVA) Process:

- 1. Identify boundary variables:** Quantity input fields have a strict minimum boundary of ≥ 1 :
 - `add_quantity`: Strict minimum ≥ 1 .
 - `edit_quantity`: Strict minimum ≥ 1 .
- 2. Identify boundary test points (3-value BVA):**
 - With a minimum boundary of $min = 1$ and a step size of 1:
 - **OFF** ($min - 1$) = 0 (Rejected or automatically normalized to 1).
 - **ON** (min) = 1 (Accepted).
 - **IN** ($min + 1$) = 2 (Accepted).
- 3. Map and supplement test cases:**
 - The OFF boundary points are already tested in domain testing (TC-CART-MOBILE-003 and TC-CART-MOBILE-009).
 - Add test cases TC-CART-MOBILE-018 and TC-CART-MOBILE-019 for `add_quantity` boundary (ON, IN).
 - Add test cases TC-CART-MOBILE-020 and TC-CART-MOBILE-021 for `edit_quantity` boundary (ON, IN).

Boundary Value Analysis Table:

Boundary B1: `add_quantity` (Closed minimum boundary ≥ 1)

Boundary Test Point	Test Value	Expected Result	Linked Test Case
OFF ($min - 1$)	0	Automatically normalized to 1	TC-CART-MOBILE-003
ON (min)	1	Accepted quantity = 1	TC-CART-MOBILE-018
IN ($min + 1$)	2	Accepted quantity = 2	TC-CART-MOBILE-019

Boundary B2: `edit_quantity` (Closed minimum boundary ≥ 1)

Boundary Test Point	Test Value	Expected Result	Linked Test Case
OFF ($min - 1$)	0	Automatically normalized to 1	TC-CART-MOBILE-009
ON (min)	1	Updated quantity = 1 in cart	TC-CART-MOBILE-020

Boundary Test Point	Test Value	Expected Result	Linked Test Case
IN (min + 1)	2	Updated quantity = 2 in cart	TC-CART-MOBILE-021

List of Test Cases Designed by BVA:

Test Case ID	Test Case Description	Tested Boundary	Boundary Point	Value	Expected Result
TC-CART-MOBILE-003	Add product from detail page with quantity = 0	B1	OFF	0	Rejected or automatically normalized to quantity 1
TC-CART-MOBILE-018	Add product from detail page with quantity = 1	B1	ON	1	Successfully added quantity 1
TC-CART-MOBILE-019	Add product from detail page with quantity = 2	B1	IN	2	Successfully added quantity 2
TC-CART-MOBILE-009	Edit quantity directly in cart to 0	B2	OFF	0	Rejected or automatically normalized to quantity 1
TC-CART-MOBILE-020	Edit quantity directly in cart to 1	B2	ON	1	Successfully updated quantity in cart to 1
TC-CART-MOBILE-021	Edit quantity directly in cart to 2	B2	IN	2	Successfully updated quantity in cart to 2

AI Gap Analysis

The Mobile Cart feature (FR-20) requires specific mobile interface testing (mobile viewport, responsive layout) and quantity synchronization. Gaps in the AI's testing for this feature include:

- **Omitting Mobile-Specific Features:** Initially, the AI designed mobile cart test scenarios identical to the desktop cart (FR-07). It missed testing elements unique to or behaving differently on mobile, such as a sticky checkout button, minimized cart drawer/modal, and swipe-to-delete behaviors.

- **Incorrect Assessment of Badge Count Logic:** The AI failed to identify the system’s logic error in calculating the cart badge quantity (the badge counted the number of unique product items instead of the sum of their quantities). The AI’s original test case merely asserted “badge updated successfully” without verifying that the badge count must equal the total quantity.
- **Lack of Mobile UI Boundary Interaction Testing:** The AI did not design test cases for directly inputting invalid characters using a mobile virtual keyboard (where users might be restricted to numeric input but can still copy-paste special characters).

Root cause: The AI lacks visual perception of mobile layouts (mobile viewports) and touch/swipe interactions. It only tested pure logical data flows, thereby missing critical bugs related to the application’s actual rendering and responsive behavior.

Bug Report

Bug ID	Title	Severity	Detecting Test Case	GitHub Link
BUG-001	[BUG][Cart Mobile] - Edit quantity directly in cart cause bad quantity counting	Major / P1	TC-CART-MOBILE-008, TC-CART-MOBILE-011, TC-CART-MOBILE-020, TC-CART-MOBILE-021	Issue #114
BUG-002	[BUG][Cart Mobile] - No confirm dialog on removing item from cart	Minor / P2	TC-CART-MOBILE-014	Issue #115
BUG-003	[BUG][Cart Mobile] - Total label not display correctly	Minor / P2	TC-CART-MOBILE-022	Issue #116
BUG-004	[BUG][Cart Mobile] - Cart Badge count the number of different items, not the total quantity	Major / P1	TC-CART-MOBILE-023	Issue #117
BUG-005	[BUG][Cart Mobile] - No illustration on empty state		TC-CART-MOBILE-024	Issue #118

Bug ID	Title	Severity	Detecting Test Case	GitHub Link
0-05		Minor / P2		

BUG-20-01: [BUG][Cart Mobile] - Edit quantity directly in cart cause bad quantity counting

- **Issue ID:** #114
- **Severity / Priority:** Major / P1
- **Test Case:** TC-CART-MOBILE-008, TC-CART-MOBILE-011, TC-CART-MOBILE-020, TC-CART-MOBILE-021
- **Environment:** Chrome, macOS, <http://localhost:8081>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Add a product to the cart on mobile.
 2. Edit quantity directly in the cart to values like 1, 2, or other valid integers.
 3. Observe the navbar badge and the quantity in the cart.
- **Expected Result:** The badge and quantity must correctly reflect the total quantity of products.
- **Actual Result:** The system miscalculates the quantity after direct edits.

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[BUG][Cart Mobile] - Edit quantity directly in cart cause bad quantity counting #114

🔗 New issue

🔗 Open

👤 yuran1811 opened 1 hour ago

👤 Owner

Found by Test Case

TC-CART-MOBILE-008, TC-CART-MOBILE-011, TC-CART-MOBILE-020, TC-CART-MOBILE-021

Requirement liên quan

FR-20

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:8081>, [85af3ba](#)

Steps to reproduce

1. Thêm sản phẩm vào giỏ trên mobile.

2. Sửa quantity trực tiếp trong giỏ ở các giá trị 1, 2 và giá trị hợp lệ khác.

3. Quan sát badge navbar và số lượng trong giỏ.

Expected result

Badge và quantity phải phản ánh đúng tổng số lượng sản phẩm.

Actual result

Hệ thống đếm sai số lượng sau khi sửa trực tiếp.

Evidence

📎 Screen.Recording.2026-06-28.at.22.11.55.mov

EShop Mobile

Đăng nhập Giỏ (2)

Giỏ Hàng

iPhone 15 Pro Max

Giá: 30,000,000 đ

Số lượng: 1

Xóa

Thành tiền: 30,000,000 đ

Samsung Galaxy S24 Ultra

Giá: 28,000,000 đ

Số lượng: 1

Xóa

Thành tiền: 28,000,000 đ

Assignees

No one - [Assign yourself](#)

Labels

module:checkout-mobile

priority:P1

severity:major

status:new

type:bug

Projects

eshop-sut

Status To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

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Evidence:

BUG-20-02: [BUG][Cart Mobile] - No confirm dialog on removing item from cart

- **Issue ID:** #115
- **Severity / Priority:** Minor / P2
- **Test Case:** TC-CART-MOBILE-014
- **Environment:** Chrome, macOS, http://localhost:8081, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Open the cart on mobile.
 2. Click the “Remove” button for a product.
- **Expected Result:** Must display a confirmation dialog before deleting.
- **Actual Result:** The product disappears immediately with no confirmation dialog.

yuran1811

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[BUG][Cart Mobile] - No confirm dialog on removing item from cart #115

New issue

Open

yuran1811 opened 1 hour ago

Owner

Found by Test Case

TC-CART-MOBILE-014

Requirement liên quan

FR-20

Severity / Priority

Minor / P2

Environment

Chrome, macOS, <http://localhost:8081>, [b5af3ba](#)

Steps to reproduce

1. Mở giỏ hàng trên mobile.

2. Bấm nút "Xóa" của một sản phẩm.

Expected result

Phải hiển thị dialog xác nhận trước khi xóa.

Actual result

Sản phẩm biến mất ngay lập tức, không có dialog xác nhận.

Evidence

ScreenRecording.2026-06-28.at.22.15.52.mov

EShop Mobile

Đăng nhậpGiỏ (2)

Giỏ Hàng

iPhone 15 Pro Max

Giá: 30,000,000 đ

Số lượng: 2

Thành tiền: 60,000,000 đ

Xóa

Samsung Galaxy S24 Ultra

Giá: 28,000,000 đ

Số lượng: 2

Thành tiền: 56,000,000 đ

Xóa

Assignees

No one - [Assign yourself](#)

Labels

module:checkout-mobile

priority:P2

severity:minor

status:new

type:bug

Projects

eshop-sut

StatusTo triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

[Create a branch](#) for this issue or link a pull request.

Notifications

Customize

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Participants

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Evidence:

BUG-20-03: [BUG][Cart Mobile] - Total label not display correctly

- **Issue ID:** #116
- **Severity / Priority:** Minor / P2
- **Test Case:** TC-CART-MOBILE-022
- **Environment:** Chrome, macOS, http://localhost:8081, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Open the cart on mobile with a valid product.
 2. Observe the price, subtotal, and total label.
- **Expected Result:** Currency display must be consistent; label must be “Tổng cộng” (Total).
- **Actual Result:** The label displays as “Tổng tạm tính” (Subtotal).

yuran1811 / hcmus-sw-testing--eshop-sut

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[BUG][Cart Mobile] - Total label not display correctly #116

New issue

Open

yuran1811 opened 1 hour ago

Owner

Found by Test Case

TC-CART-MOBILE-022

Requirement liên quan

FR-20

Severity / Priority

Minor / P2

Environment

Chrome, macOS, http://localhost:8081, 85af3ba

Steps to reproduce

1. Mở giỏ hàng trên mobile với sản phẩm hợp lệ.

2. Quan sát giá, thành tiền và nhãn tổng.

Expected result

Tiền tệ hiển thị nhất quán; nhãn phải là "Tổng cộng".

Actual result

Nhãn hiển thị là "Tổng tạm tính".

Evidence

EShop Mobile

Đăng nhập Giỏ (2)

Giỏ Hàng

iPhone 15 Pro Max

Giá: 30,000,000 đ

Số lượng: 1

Xóa

Thành tiền: 30,000,000 đ

Samsung Galaxy S24 Ultra

Giá: 28,000,000 đ

Số lượng: 1

Xóa

Thành tiền: 28,000,000 đ

Tổng tạm tính: 58,000,000 đ

Assignees

No one - [Assign yourself](#)

Labels

module:checkout-mobile priority:P2

severity:minor status:new type:bug

Projects

eshop-sut

Status To triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

Create a branch for this issue or link a pull request.

Notifications

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Participants

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Evidence:

BUG-20-04: [BUG][Cart Mobile] - Cart Badge count the number of different items, not the total quantity

- **Issue ID:** #117
- **Severity / Priority:** Major / P1
- **Test Case:** TC-CART-MOBILE-023
- **Environment:** Chrome, macOS, <http://localhost:8081>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Add 1 product with quantity = 2.
 2. Add another product with quantity = 1.
 3. Observe the navbar badge.
- **Expected Result:** The badge must display the total quantity of all items.
- **Actual Result:** The badge counts unique product items instead of total quantity.

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Q Type to search

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Issues 105

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[BUG][Cart Mobile] - Cart Badge count the number of different items, not the total quantity #117

New issue

Open

yuran1811 opened 1 hour ago

Owner

Found by Test Case

TC-CART-MOBILE-023

Requirement liên quan

FR-20

Severity / Priority

Major / P1

Environment

Chrome, macOS, <http://localhost:8081>, [B5af3ba](#)

Steps to reproduce

1. Thêm 1 sản phẩm với quantity = 2.

2. Thêm thêm sản phẩm khác quantity = 1.

3. Quan sát badge navbar.

Expected result

Badge phải hiển thị tổng quantity của các item.

Actual result

Badge đếm theo số dòng sản phẩm thay vì tổng số lượng.

Evidence

EShop Mobile

Đăng nhậpGiỏ (2)

Giỏ Hàng

iPhone 15 Pro Max

Giá: 30,000,000 đ

Số lượng: 3

Thành tiền: 90,000,000 đ

Xóa

Samsung Galaxy S24 Ultra

Giá: 28,000,000 đ

Số lượng: 1

Thành tiền: 28,000,000 đ

Xóa

Assignees

No one - [Assign yourself](#)

Labels

module:checkout-mobile

priority:P1

severity:major

status:new

type:bug

Projects

eshop-sut

StatusTo triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

[Create a branch](#) for this issue or link a pull request.

Notifications

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Participants

Transfer issue

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Give feedback

Evidence:

BUG-20-05: [BUG][Cart Mobile] - No illustration on empty state

- **Issue ID:** #118
- **Severity / Priority:** Minor / P2
- **Test Case:** TC-CART-MOBILE-024
- **Environment:** Chrome, macOS, <http://localhost:8081>, 85af3ba875c88283615e22cb108f13e2fccaf0e9
- **Steps to Reproduce:**
 1. Open an empty cart on mobile.
 2. Observe the empty state area.
- **Expected Result:** The empty state must feature an illustration or icon.
- **Actual Result:** Only text and a link are present, with no illustration.

hcmus-sw-testing--eshop-sut

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[BUG][Cart Mobile] - No illustration on empty state #118

New issue

Open

yuran1811 opened 1 hour ago

Owner

Found by Test Case

TC-CART-MOBILE-024

Requirement liên quan

FR-20

Severity / Priority

Minor / P2

Environment

Chrome, macOS, http://localhost:8081, 85af3ba

Steps to reproduce

1. Mở giỏ hàng trống trên mobile.
2. Quan sát khu vực empty state.

Expected result

Empty state phải có hình minh họa hoặc icon.

Actual result

Chỉ có text và link, không có hình minh họa.

Evidence

EShop Mobile

Đăng nhậpGiỏ (0)

Giỏ Hàng

Giỏ hàng của bạn đang trống

Tiếp tục mua sắm

Create sub-issue

Assignees

No one - Assign yourself

Labels

module:checkout-mobilepriority:P2severity:minorstatus:newtype:bug

Projects

eshop-sut

StatusTo triage

Milestone

hw2

Due by June 29, 2026

Relationships

None yet

Development

Create a branch for this issue or link a pull request.

Notifications

Customize

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Participants

Transfer issue

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yuran1811 added this to the hw2 milestone 1 hour ago

Evidence: